CASE STUDY: SIMOBY CRM & DATA SILO INTEGRATION

Role:	Project Manager & Contributing Developer (Student Employee)
Department:	UNY Office of International Services
Framework:	Yii Framework (PHP)
Live URL:	https://simoby.uny.ac.id

1. Project Overview

SIMOBY (Sistem Informasi Mobilitas Internasional UNY) is a centralized CRM and workflow automation platform engineered for the Office of International Services. Serving as a Student Employee at the International Office, I acted as the primary Project Manager for this system. I bridged the gap between our department's complex bureaucratic needs and the central UNY IT Development Team.

2. Key Challenges & Contributions

A. Cross-Functional Collaboration & Requirement Translation

The Challenge: University data was heavily fragmented across legacy systems. Staff identity data resided in the HR portal (SIAP), while student academic records were locked inside the Academic Information System (SIAKAD). The challenge was translating these rigid administrative rules into clear technical specifications for the engineering team.

The Contribution: I managed the project lifecycle and collaborated closely with the central UNY IT Team to design an integration layer. I ensured the final architecture successfully synchronized validated user records via API, effectively eliminating manual data entry and breaking down institutional data silos.

B. Managing Multi-Step State Workflows & Code Contributions

The Challenge: An overseas mobility application is highly relational, requiring multi-source budgeting tracking and strict state-machine approvals (Submission → Verification → Funding → Finalization → Reporting).

The Contribution: While the core Yii framework architecture was spearheaded by the senior IT team, I actively contributed to the codebase by implementing minor features, conducting extensive QA testing, and

managing the system's rollout. I also authored the official User Manual and digitalized the entire workflow process.

3. System Workflow Architecture

The infographic below details the multi-step state machine and approval workflow I helped design and document to simplify the complex bureaucratic processes.

PROSES PENGAJUAN IZIN PERJALANAN DINAS LUAR NEGERI (PDLN) MELALUI SIMOBY

1 **Login simoby**
Gunakan akun SSO.



2 **Pengajuan kegiatan**
Pilih kegiatan yang tersedia. Tambahkan dengan kegiatan baru jika belum ada kegiatan.



3 **Data identitas diri**
Upload dokumen yang diperlukan (KTP, kartu Pegawai, LoI, Rundown Kegiatan, CV, Paspor, dan SK Pangkat Terakhir).



4 **Pembiayaan**
Input data pembiayaan perjalanan dinas sesuai sumber pendanaan: Mandiri/Universitas/Sponsor



5 **Upload dokumen keberangkatan**
Dokumen yang diberi tanda bintang (*) merupakan dokumen wajib (LoA, Visa, Paspor, dll).



6 **Finalisasi dan konfirmasi**
Cek aplikasi data yang dilampirkan sudah sesuai dan lengkap. Jika sudah klik Submit.



7 **Penerbitan dokumen keberangkatan**
Melalui aplikasi SIMPEL Setneg.



8 **Laporan**
Buat laporan akademis dan keuangan perjalanan PDLN yang telah dilaksanakan dan diunggah Simoby.



Pengajuan tidak boleh kurang dari 2 bulan sebelum keberangkatan!
Usulan Perjalanan tidak akan diproses sebelum laporan perjalanan dinas sebelumnya terunggah!